



Smart Water Usage Advisor

Architecting Sustainability for UN SDG 6

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Defining the Crisis



2.2 Billion

The global population currently lacking access to safely managed drinking water services.



Institutional Waste

Large campuses consume thousands of liters unnecessarily due to behavioral inertia and leak ignorance.



Knowledge Gap

Static municipal reports fail to engage modern students in day-to-day conservation habits.

SDG 6: Alignment & Goals

Target 6.4

Substantially increase water-use efficiency across all sectors and ensure sustainable freshwater withdrawals to address water scarcity.

The AI Solution

By leveraging AI for real-time mentorship, we empower individuals to achieve global sustainability targets through local action.

Core Solution Architecture



24/7 AI Mentorship: A conversational advisor powered by IBM BOB that provides instant conservation tips tailored for hostel life.



Impact Calculator: A behavioral tracking tool that quantifies water savings in liters based on daily user inputs.



Simulated RAG Pipeline: Demonstrates document-driven responses by processing municipal water guideline PDFs.

Responsible AI Framework: Integrated disclaimers to ensure transparency and adherence to local legal policies.

Technical Infrastructure



IBM BOB

Used for prompt engineering and conversational logic orchestration.



Web Stack

Built with HTML5, CSS3, and JavaScript for a responsive, water-themed UI.



GH Pages

Hosted on GitHub for open accessibility and version control management.

| Quantifying Conservation

15L

Per Bucket Saved

20L

Per Leak Fixed

45L

Daily Avg. Target

The interactive calculator gamifies sustainability by showing users exactly how much they contribute to the campus-wide water conservation goal every single day.

Water Consumption Analysis

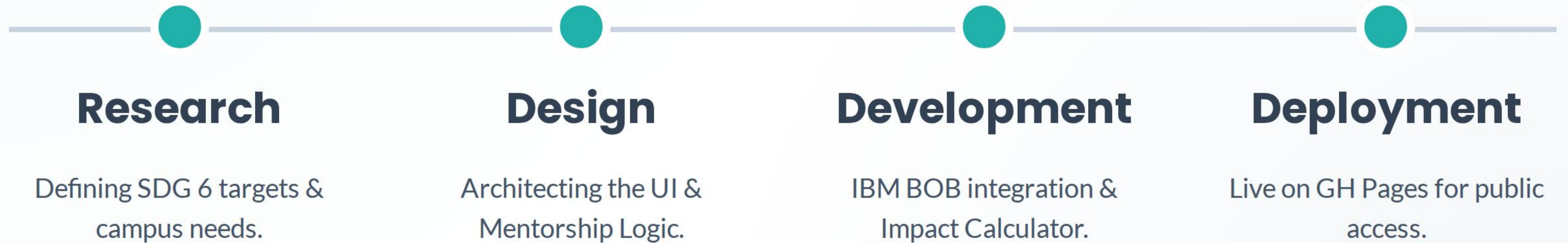
Activity	Standard Usage	With AI Advisor Tips	Savings (%)
Dorm Showering	60 Liters	35 Liters	42%
Personal Hygiene	15 Liters	8 Liters	47%
Laundry (Manual)	40 Liters	25 Liters	37%

Engagement Comparison



Interactive AI mentorship significantly outperforms traditional awareness methods by providing real-time, actionable value to the student community.

Project Evolution



Innovator's Vision

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"Water is the lifeblood of our planet. By combining AI with human conscience, we can turn every student into a steward of our most precious resource."

— Yogamoorthy

Scalability & Future Scope

IoT Integration

Connecting the AI advisor to smart water meters for automated, real-time consumption feedback and leak alerts.

Institutional Policy

Expanding the RAG engine to process diverse university sustainability policies across the globe.



Questions?

Thank you for your time and for supporting sustainability.

Yogamoorthy

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